

FROM PROJECT TO PRODUCT

Customer Review Sentiment Analytics

Concepts explained simply · pitch scripts · how to turn this project into a business

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Companion to the project report and the presentation script

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How to use this guide

The project already comes with a **project report** (what was built and how it was tested) and a **presentation script** (what to say in the demo and viva). This third document does two jobs:

- **Part A: Concepts.** Every idea in the project explained in plain English with an analogy, where it appears in the project, and the one-line answer to give an examiner.
- **Part B: Pitch scripts.** Ready-to-say scripts at 30 seconds, 2 minutes and 5 minutes, for an examiner, for a client, and for an investor or mentor.
- **Part C: Business.** Who would pay for this, what to sell, what to build next, pricing ideas, how to find the first customers, legal points, risks and a 90-day launch plan.

Honest status. The project is a strong, working **prototype**: a trained model, an interactive dashboard and tests. It is not yet a finished product. Part C is explicit about what is missing before you charge anyone, and every price or cost figure in it is an **assumption for planning** that you must check with real quotes, not a measured fact.

Part A: Concepts Explained Simply

Each concept follows the same pattern: **what it is**, **an everyday analogy**, **where it appears in this project** and **what to say if asked**.

1. Big data and the five Vs

What it is	Data so large, fast, varied or messy that ordinary tools struggle. Described by Volume, Velocity, Variety, Veracity and Value.
Analogy	A library that gets a thousand new books every day, in every language, some with pages torn out. You need a system, not a person, to make sense of it.
In this project	199,426 reviews, a matrix with 150,708 columns, noisy text, noisy star labels, and the business value of finding what customers dislike.
Say this	"It is big-data because of the size of the text features and the way we handle them: sparse storage, a scalability study and streaming training."

2. Sentiment analysis

What it is	Teaching a computer to decide whether a piece of text is positive, negative or neutral.
Analogy	A shop assistant who can tell from a customer's words whether they walked out happy, angry or indifferent.
In this project	The whole project: every review gets one of three labels plus a probability.
Say this	"It turns thousands of opinions into numbers a manager can act on."

3. Supervised machine learning

What it is	Learning from examples that already have the right answer (labels), then predicting the answer for new examples.
Analogy	Learning to recognise fruit by being shown 100,000 labelled photos, then naming a fruit you have never seen.
In this project	We learn from reviews whose star rating gives the label, then predict for unseen reviews.
Say this	"Stars are our labels; the model learns which words go with which label."

4. Features and TF-IDF

What it is	Computers need numbers, so each review becomes a long list of numbers: one per word. TF-IDF makes a word's number high when it is frequent in this review but rare in all reviews.
Analogy	Highlighting the unusual words in a document: 'the' appears everywhere so it is ignored; 'defective' is rare so it stands out.
In this project	TF-IDF over 150,708 words and word pairs (src/modeling.py).

Say this	"TF-IDF weights words by how distinctive they are, so common words do not drown out meaningful ones."
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5. n-grams (bigrams)

What it is	Sequences of 2 or more words kept together, such as 'not good' or 'stopped working'.
Analogy	Reading 'ice cream' as one thing, not as 'ice' and 'cream' separately.
In this project	Unigrams plus bigrams. The ablation shows about +2 points macro-F1 over single words.
Say this	"Bigrams let a simple model understand short phrases and negation."

6. Stop-words and negation

What it is	Stop-words are very common words ('the', 'and') that carry little meaning and are usually removed. Negations ('not', 'never') must be kept because they flip the meaning.
Analogy	Removing 'not' from 'not good' leaves 'good', like deleting the minus sign from -5.
In this project	Custom stop-word list that keeps negations and intensifiers. Removing them costs about 2.6 points of macro-F1.
Say this	"A standard stop-word list would turn 'not good' into 'good', so we built our own."

7. Sparse matrix

What it is	A table that remembers only the cells that are not zero. Text tables are almost entirely zeros.
Analogy	A seating plan that lists only occupied seats in a 50,000-seat stadium instead of every empty one.
In this project	The test matrix is 99.985% zeros: 7 MB sparse versus 24 GB dense.
Say this	"Sparse storage is why this fits on a laptop."

8. Logistic Regression (the classifier)

What it is	Gives each word a weight for each class, adds up the weights present in a review, and converts the three totals into probabilities (softmax).
Analogy	A judge who awards points: 'terrible' +3 for negative, 'love' +3 for positive, then announces who has the most points and how sure they are.
In this project	The selected model. Because each word has a visible weight, we can explain every decision.
Say this	"It is fast, accurate enough, gives probabilities and is explainable."

9. Other models we compared

What it is	Naive Bayes (counts words, assumes independence), SVM (draws the widest possible dividing line), SGD (learns one example at a time), baseline (always guess the most common class).
Analogy	Different judges with different styles. The baseline is the judge who always says 'positive'.
In this project	Six models compared on validation. Linear models won; the baseline scored only 40 %.
Say this	"We did not assume the first model was best; we compared six."

10. Class imbalance

What it is	When one class is much smaller. Here neutral is 20 % of the data, so a lazy model could ignore it.
Analogy	A fire alarm that never rings is right 99.9 % of the time but useless.
In this project	Balanced class weights (neutral counts about twice as much) and macro-F1 as the headline metric.
Say this	"We weighted the small class and judged by macro-F1 so it cannot be ignored."

11. Train, validation and test sets

What it is	Three separate piles of data. Train: learn. Validation: choose and tune. Test: final exam, used once.
Analogy	Studying from a textbook (train), taking practice papers to decide how to revise (validation), then the real exam (test).
In this project	A stratified 70/10/20 split; the test set was touched once.
Say this	"The test number is honest because the model never saw those reviews and we never tuned on them."

12. Overfitting and regularisation

What it is	Overfitting is memorising the training data instead of learning general patterns. Regularisation (the parameter C) keeps the model simple to prevent it.
Analogy	A student who memorises last year's answers and fails when the questions change.
In this project	C was tuned on validation; validation and test scores are close, so there is no sign of overfitting. Chosen C = 0.5.
Say this	"Validation and test scores agree, so it generalises."

13. Data leakage

What it is	Accidentally letting information from the test data influence training, making results look better than they are.
Analogy	Seeing the exam paper before the exam.
In this project	Vocabulary fitted on training data only; grid search re-fits the vectoriser inside each fold; test reviews excluded from the stream.
Say this	"We fit every transformation on training data only."

14. Accuracy, precision, recall, F1

What it is	Accuracy: share of all predictions that are right. Precision: of the reviews we called X, how many really are X. Recall: of the real X reviews, how many did we find. F1: a single number combining precision and recall.
Analogy	A fisherman: recall is how many fish in the lake he catches; precision is how many of his catch are the right species.
In this project	Accuracy 71.2%; negative F1 0.76, neutral 0.45, positive 0.81.
Say this	"Accuracy alone hides weak classes, so we report per-class numbers too."

15. Macro-F1

What it is	The simple average of the F1 of each class, so every class counts equally regardless of its size.
Analogy	Grading a school by the average of each class's marks, not by the biggest class alone.
In this project	Macro-F1 = 0.676, our headline metric.
Say this	"It is honest about the weak neutral class."

16. Confusion matrix

What it is	A table of actual versus predicted classes that shows which classes are mistaken for which.
Analogy	A report of which twins a teacher keeps mixing up.
In this project	Neutral reviews are confused with both other classes; positive and negative are rarely confused with each other.
Say this	"It shows the model's mistakes are mostly neutral confusions, not opposite-sentiment errors."

17. ROC-AUC

What it is	How well the model ranks examples: the chance that a random positive example gets a higher score than a random negative one. 0.5 is chance, 1.0 is perfect.
Analogy	Sorting a pile of exam papers and checking how often a good paper lands above a bad one.
In this project	Macro AUC 0.862; neutral is the lowest at 0.76.
Say this	"AUC does not depend on the decision threshold."

18. Explainability

What it is	Showing why a model made a decision, in terms a human can check.
Analogy	A doctor explaining the symptoms behind a diagnosis rather than just announcing it.
In this project	The dashboard highlights each word by the class it supports and charts the strongest terms.
Say this	"Contribution = TF-IDF value times the word's weight for the predicted class."

19. Scalability, out-of-core learning and hashing

What it is	Scalability: does it still work when data grows? Out-of-core: learn from data in chunks without loading everything. Hashing trick: map each word to a bucket number so no vocabulary list is needed.
Analogy	Reading a huge book one chapter at a time, keeping only your notes, instead of memorising the whole book at once.
In this project	Learning curve and timing study; streaming training with HashingVectorizer and <code>partial_fit</code> ; mapping to Spark.
Say this	"Cost grows roughly linearly and the model can learn in chunks, so it can grow with the data."

20. Why the order of a data stream matters

What it is	Online learners assume examples arrive in random order. If a file is sorted by label, the model only remembers the last labels it saw.
Analogy	Revising only the last chapter before an exam covering the whole book.
In this project	Our raw file is sorted by star rating; a naive pass scored about 40 %, an out-of-core shuffle fixed it. This is a genuine finding from the project.
Say this	"We checked the order of the stream, found the problem and fixed it."

21. Dashboard and deployment

What it is	A dashboard is an interactive page where non-programmers use the model. Deployment means making it available to others.
Analogy	The difference between a recipe and a restaurant.
In this project	Streamlit app with six pages; saved model loaded instantly; batch CSV upload and download.
Say this	"It turns the model into a tool that a manager can use without code."

Worked example: reading the confusion matrix

Using the real test results, the table below shows what the model did with the actual reviews in each class.

Table 1: Confusion matrix on the test set

Actual class	Predicted negative	Predicted neutral	Predicted positive	Total	Recall
Negative	11,778	3,414	772	15,964	73.8%
Neutral	2,383	4,066	1,533	7,982	50.9%
Positive	888	2,506	12,546	15,940	78.7%

- **Recall for neutral** = $4,066 / 7,982 = 50.9\%$: of all truly neutral reviews, about half were found.
- **Precision for neutral** = $4,066 / 9,986 = 40.7\%$: of everything labelled neutral, only about 41% really was neutral. Many positive and negative reviews were pulled into the middle.
- **Overall accuracy** = $(11,778 + 4,066 + 12,546) / 39,886 = 71.2\%$.
- Notice that opposite-sentiment mistakes (negative predicted as positive and vice versa) are rare: the model's errors are mostly 'close' mistakes involving neutral.

Part B: Pitch Scripts

Three audiences, three levels of detail. Speak the ideas, do not read the words. Numbers come from the measured results of the project.

B1. For the examiner (30 seconds)

"Our project classifies about 200,000 Amazon reviews as positive, neutral or negative with 71% accuracy, against 40 % for guessing. We compared six models, tuned only on a validation set, and tested once on 39,886 unseen reviews. It is explainable, scales by streaming, and is delivered as a notebook and an interactive dashboard."

B2. For the examiner (2 minutes)

1. **Problem.** "Stores get far more reviews than anyone can read, and star ratings do not say why customers are unhappy."
2. **Data.** "We used 199,426 real Amazon reviews; 1-2 stars became negative, 3 neutral, 4-5 positive."
3. **Method.** "We clean the text but deliberately keep words like 'not'. We convert reviews to TF-IDF numbers over words and word pairs, and compare six models."
4. **Result.** "Logistic Regression is best: 71.2% accuracy, 0.68 macro-F1, AUC 0.86. Positive and negative are recognised well; neutral is hardest because 3-star reviews are ambiguous."
5. **Big data.** "The matrix is almost all zeros so we store it sparsely; accuracy keeps rising with more data; and we can train on the file in chunks without loading it all. We even found the file was sorted by rating, which broke naive streaming, and fixed it."
6. **Delivery.** "A dashboard lets anyone paste a review, see the prediction and the words behind it, or upload a CSV."
7. **Honesty.** "Star labels are noisy and the model cannot detect sarcasm. A transformer and aspect-based sentiment are our next steps."

B3. For a business owner (60 seconds, no jargon)

"You have thousands of customer reviews and no time to read them. Our tool reads every review in seconds, tells you how many customers are happy, neutral or unhappy, and shows you *what* the unhappy ones talk about, such as delivery, product quality or customer support. You upload a spreadsheet and get a report and a dashboard. You find your biggest problems in minutes instead of weeks, and you can track whether things improve after you fix them."

B4. Investor or mentor pitch (about 3 minutes)

Table 2: Investor pitch outline

Slide / beat	What to say
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Slide / beat	What to say
1. Problem	"Online sellers and service businesses are drowning in reviews. They can see a 3.8-star average but not the reasons behind it, and they find out about problems too late."
2. Solution	"We turn raw reviews into a clear sentiment score and a ranked list of complaint themes, with a dashboard that explains every decision."
3. Proof	"A working prototype: trained on about 200,000 real reviews, 71% accuracy against a 40 % baseline, tested on 39,886 unseen reviews, with automated tests and a live demo."
4. Who pays	"Small and mid-size online sellers, restaurants and hotels, app developers and agencies who manage reviews for several clients."
5. Product	"A web dashboard first, then an API and monthly insight reports; later, aspect-level insights and alerts."
6. Business model	"Monthly subscription by review volume, plus one-off analysis projects for businesses that only need a report."
7. Differentiation	"Affordable, explainable, tuned to the customer's own data rather than a generic score, and simple enough for a non-technical owner."
8. Plan and ask	"In 90 days: validate with 5 pilot businesses on their own data, harden the product, then charge. We are looking for pilot customers and guidance."

Part C: Turning the Project into a Business

C1. The problem you are selling a solution to

Reviews decide buying decisions, ranking and reputation, yet most small businesses read only the latest few. The pain is specific and measurable:

- They do not know **why** customers are unhappy, only that the average dropped.
- They find out about a faulty batch, a slow courier or a rude support agent weeks late.
- Agencies and marketplace sellers with many products cannot read reviews for all of them.
- In this project's data, among reviews that mention customer service or refunds, 70% are negative versus 40 % overall: **the reasons are findable, but only if someone analyses the text.**

C2. Who would pay

Table 3: Customer segments

Segment	Pain	What they would buy	Why they might pay
Online sellers (own website, marketplace, Instagram shops)	Hundreds to thousands of reviews, no time to read.	Monthly sentiment report + dashboard.	Protects ratings and sales; finds product faults early.
Restaurants, cafes, hotels, clinics	Reviews on Google/Zomato/TripAdvisor-style sites.	Weekly summary of complaints by theme (food, service, cleanliness).	Reputation drives walk-in business.
Mobile app and SaaS companies	App-store reviews and support tickets.	Sentiment trend + alerts after each release.	Spot a bad release within hours.
Marketing and e-commerce agencies	Report on many clients' brands.	White-label dashboard or API.	Saves analyst hours across clients.
Customer-support teams	Hard to prioritise unhappy customers.	Batch triage: negative first, with confidence.	Faster response to angry customers.

Start narrow. Pick one segment and one channel (for example, small online sellers) and solve their problem very well before widening. The easiest first customers are people you can reach personally and who will give you their reviews for a free pilot.

C3. What to sell (offers from easiest to hardest)

Table 4: Offers

Offer	What the customer gets	What you need to build	Effort
1. One-off review audit (service)	A report on their last 500-5,000 reviews: sentiment mix, top complaint themes, example reviews, 5 recommendations.	Nothing new: use the current notebook and dashboard with their CSV.	Low (start here)
2. Monthly insight report	Same report each month with trends and changes.	A repeatable script and report template.	Low-medium

Offer	What the customer gets	What you need to build	Effort
3. Hosted dashboard (subscription)	Login, upload or connect reviews, see sentiment and themes, download results.	Authentication, database, multi-tenant storage, hosting.	Medium
4. API	Send review text, receive sentiment and confidence.	A REST API (for example FastAPI), API keys, rate limits, docs.	Medium
5. White-label / custom model	Model trained on the client's own labelled data, branded for agencies.	Training pipeline per client, evaluation and monitoring.	High

C4. What is missing before you charge (honest gap list)

Table 5: Gaps to close

Gap	Why it matters	How to close it
Model was trained on Amazon product reviews	Words differ by domain: 'cheap' or 'hot' mean different things for a phone, a restaurant or a hotel. Accuracy on a new domain will be lower than 71 %.	Validate on 200-500 of the client's own reviews labelled by hand; retrain or fine-tune per client.
Dataset licence	The Amazon review corpus is provided for research and education. A commercial service should not rely on a model trained on data you are not licensed to use commercially.	Train the production model on data you own or are licensed to use (clients' own reviews, with permission, or a commercially licensed dataset). Read the licence terms.
Neutral class is weak (F1 about 0.45)	A business may care most about the middle ground.	Offer positive/negative with a confidence threshold ('needs review' band), or collect better neutral labels.
English only	Many local businesses get Hindi, Marathi or Hinglish reviews.	Add a multilingual model (for example a multilingual transformer) in version 2.
No sarcasm or aspect detection	'Great, it broke in a day' is read as positive; owners want to know what each review is about.	Add aspect-based sentiment and a transformer model.
No login, database or API	A subscription product needs accounts, storage and integrations.	Add FastAPI + PostgreSQL + authentication; deploy with Docker.
No monitoring	Language drifts and accuracy can decay silently.	Track accuracy on a small labelled sample each month; keep a feedback button.

C5. Product roadmap

Table 6: Roadmap

Stage	Time	Deliverables
MVP (service)	Weeks 1-4	Review audit service using the existing project; report template; 3-5 free pilots on real client data; measure accuracy on their own labelled sample.
v1 (product)	Months 2-3	Hosted dashboard with login; CSV upload; saved history; theme discovery; PDF/Excel export; payment link.

Stage	Time	Deliverables
v2	Months 4-6	API; connectors/exports for common review sources; Hindi/Hinglish support; alerts when negative share jumps.
v3	Months 7-12	Aspect-based sentiment; transformer model; per-client fine-tuning; Slack/email digests; team accounts.

C6. Production architecture (what to add to the prototype)

Table 7: From prototype to production

Component	Prototype today	For a real product
Interface	Streamlit app	Keep Streamlit for internal use; web front-end or Streamlit behind authentication for clients.
Model serving	Joblib file loaded in the app	FastAPI service wrapping the model; Docker container; health checks.
Storage	Local files	PostgreSQL for accounts, uploads and results; object storage for files.
Data ingestion	CSV upload	Scheduled imports from the sources the client is permitted to export; webhook endpoint.
Jobs	Run on demand	Background worker for large uploads (queue); retraining job.
Monitoring	Tests only	Logging, error alerts, accuracy checks on a labelled sample, model versioning.
Security	Local only	HTTPS, hashed passwords, per-tenant data isolation, backups, data-retention rules.

C7. Business model and pricing (illustrative assumptions)

These numbers are planning assumptions, not market facts. Test them by asking 10 prospective customers what they would pay, and compare with what similar tools charge.

Table 8: Pricing ideas (assumptions)

Plan	Suggested price (assumption)	Included	Target customer
Audit (one-off)	Rs 3,000 - 15,000 per project	Report on up to 5,000 reviews, 30-minute walkthrough, recommendations.	Small sellers, restaurants
Starter	Rs 999 - 1,999 per month	Up to 2,000 reviews/month, dashboard, CSV export.	Solo sellers
Growth	Rs 4,999 - 9,999 per month	Up to 20,000 reviews/month, themes, trends, alerts, email support.	Growing brands, agencies
API / white-label	Usage-based or custom quote	API keys, higher limits, branding, custom model.	Agencies, SaaS firms

Rough cost picture (to verify with real quotes)

- **Hosting:** a small cloud server or container service is typically a low monthly cost at the start; get an actual quote for your chosen provider and region.
- **Compute for predictions is tiny:** the model predicts about 7,323 reviews per second on a laptop CPU and the model file is 5 MB, so no GPU is needed for the current approach.
- **Your time is the main cost** in the first months: sales, onboarding and labelling a validation sample for each new client.
- **Fees to remember:** payment-gateway fees, domain name, email service, taxes (check GST applicability for services) and business registration.

C8. How to find your first 10 customers

1. **List 50 businesses you can reach personally** (friends' and family businesses, local shops with online presence, college-event sellers, freelancers).
2. **Offer a free pilot:** "Give me your last 300 reviews; I will send a one-page insight report in 48 hours."
3. **Use the demo:** screen-record a 2-minute video of the dashboard; share it on LinkedIn with an anonymised pilot result.
4. **Turn each pilot into a case study** (with permission): problem, what the analysis found, what the owner changed, the result.
5. **Convert:** after the free report, offer the monthly report or the Starter plan.
6. **Freelance platforms:** package the audit as a fixed-price gig; reviews and ratings build credibility.
7. **Partnerships:** agencies and consultants who already serve small sellers can resell or refer.

C9. Sample deliverable for a client (what the report contains)

Table 9: Client report outline

Section	Content	Source in this project
1. Snapshot	Number of reviews, sentiment mix, comparison with their star average.	Batch analysis page
2. Themes	Which topics drive negative reviews (delivery, quality, support ...) with percentages.	Theme analysis (notebook section 11)
3. Phrases	Most common positive and negative phrases.	Data Insights page
4. Evidence	10 typical negative reviews and 10 typical positive reviews, with the key words highlighted.	Analyze page highlighting
5. Recommendations	Top 3-5 actions ordered by impact.	Analyst judgement
6. Method and limits	How accurate the analysis is on a sample of their own reviews.	Evaluation approach in the report

C10. Competition and differentiation

Large cloud providers (such as Google Cloud Natural Language, Amazon Comprehend and Microsoft Azure AI Language) and specialised review-analytics vendors already offer sentiment analysis, so you will not win on the algorithm alone. Realistic differentiators for a small team:

- **Price and simplicity:** a non-technical owner uploads a sheet and gets a plain-language report; no cloud accounts or code.
- **Explainability:** word-level evidence for every prediction builds trust.
- **Fit to the customer's own data:** a short calibration on their reviews often beats a generic score.
- **Local focus:** Hindi, Marathi and Hinglish support, and local review sources, are underserved.
- **Human service:** a short call that turns the numbers into actions.

Compare current features and prices of these competitors yourself before you finalise your positioning; offerings change.

C11. Legal, ethical and compliance points

- **Dataset licence.** Do not sell a product built on the Amazon review dataset without checking its terms; it is intended for research and education. Use client-owned or properly licensed data for the production model.
- **Do not scrape against a site's terms.** Use official exports, APIs or data the client provides. Respect each platform's terms of service.
- **Personal data.** Reviews can contain names or other personal details. In India, consider obligations under the Digital Personal Data Protection Act, 2023; for EU users, GDPR. Collect only what you need, state how long you keep it, and let clients delete data.
- **Contracts.** Use a simple written agreement covering what you deliver, who owns the data and results, confidentiality and a limitation of liability. Get advice from a lawyer for the template.
- **Honest claims.** Report accuracy measured on the client's own sample, and describe limits (sarcasm, neutral reviews). Never claim 100 %.
- **Fair use of results.** Sentiment scores must not be used to automatically punish individual reviewers or to hide legitimate criticism.
- **Business registration and tax.** Check requirements for freelancing or registering a company where you live before invoicing.

C12. Risks and how to reduce them

Table 10: Risks and mitigations

Risk	Likelihood / impact	Mitigation
Accuracy drops on a new domain	High / High	Validate on 200-500 client reviews before promising numbers; calibrate or retrain per client.
Customers do not see the value	Medium / High	Lead with decisions and actions, not charts; show one surprising finding in the free pilot.
Big vendors offer the same cheaply	Medium / Medium	Niche focus, explainability, local languages, personal service.
Data licence or privacy problem	Medium / High	Use only permitted data; written agreements; minimal retention.
Model decays over time	Medium / Medium	Monthly accuracy check on a labelled sample; feedback loop.
You are one person	High / Medium	Keep scope small; automate report generation; use templates; say no to custom work early.

C13. A 90-day launch plan

Table 11: 90-day plan

Weeks	Goal	Concrete actions
1-2	Package the service	Write a one-page offer; build the report template from the notebook; record a 2-minute demo video; prepare a short contract and privacy note.
3-4	First pilots	Approach 20 businesses; complete 3-5 free pilots; label 200 reviews per pilot to measure real accuracy; collect feedback.
5-6	Learn and fix	Fix vocabulary gaps found in pilots; add domain-specific stop-words or retrain; improve the report.
7-8	First paid offers	Convert two or three pilots into paid audits; publish one case study.
9-10	Productise	Add login and storage; automate the monthly report; set up a payment link.
11-12	Review and decide	Measure: pilots completed, conversion rate, accuracy on client data, hours per report. Decide whether to build the API or deepen the service.

C14. Numbers to track

Table 12: Metrics

Metric	Why it matters
Pilots started / completed	Shows demand and your delivery speed.
Accuracy on the client's own labelled sample	The credibility of every claim.
Conversion from free pilot to paid	The best early signal that the product has value.
Hours spent per report	Tells you what to automate first.
Monthly recurring revenue and churn	Health of a subscription business.
Customer-reported value (decisions made, issues fixed)	Your strongest marketing material.

C15. Using the project for your career

- **Portfolio:** the GitHub repository, the executed notebook and the live demo show end-to-end skills (data, modelling, evaluation, app, tests).
- **Freelancing:** the review-audit service uses what you already built; start with a fixed-price gig.
- **Interviews:** be ready to explain the neutral-class problem, the ablation study, the leakage precautions and the sorted-stream discovery; they show real judgement.
- **Next skills to add:** FastAPI, Docker, SQL, a transformer fine-tune and basic cloud deployment, which together turn a prototype into a deployable product.

C16. Your next three actions

1. Choose one segment (for example, small online sellers) and write down 20 specific businesses you can contact this week.
2. Prepare the one-page offer and run the first free pilot using `data/demo_reviews.csv` as a rehearsal.

3. Label 200 reviews from the pilot and measure the model's accuracy on them before you quote any number to a client.